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A Precise Calculation of the Critical Rayleigh Number and Wave Number for the Rigid-Free Rayleigh-Bénard Problem

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Abstract

The critical Rayleigh number R_c and critical wave number k_c arise as threshold constants in the consideration of the planar Rayleigh-Bénard problem. The purpose of this short communication is to announce the calculation of error-bounded, verifiable estimates for R_c and k_c for the asymmetric 'rigid-free' boundary condition formulation of the problem. Numerical values given are guaranteed accurate to fifty decimal places and are compared to calculations in the literature.

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1 Introduction

In 1900, Henri Bénard [2] documented the formation of cellular structure formed within a thin layer of fluid heated from below. Since then, Rayleigh-Bénard convection has been the subject of intense study, in large part due to

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the interplay between the physical simplicity and mathematical complexity of the problem. Lord Rayleigh's 1916 paper [21] was the first mathematical treatment of the problem; subsequent work appeared over the next fifty years from Jeffreys [13], Low [17], Pellew and Southwell [20], Reid and Harris [22] and Chandrasekhar [7]. Since then, advances in nonlinear analysis have spurred literally hundreds of further papers. Several important and more modern contributions can be found in [3, 4, 6, 8, 9, 10, 12, 15, 16, 19] and elsewhere.

2 Mathematical Preliminaries

Heating the bottom surface of a layer of fluid creates an adverse temperature gradient, and hotter fluid in the bottom of the layer expands. When this increased buoyancy is sufficient to overcome the viscosity and heat conductivity of the fluid, convection sets in. In [21], Lord Rayleigh analyzed the problem via the Boussinesq approximation to the Navier-Stokes equations for a fluid assumed incompressible except for thermal buoyancy [5]. He predicted that the onset of convection is determined by the parameter R, now referred to as the *Rayleigh number*:

$$R = \frac{g\alpha}{\kappa\nu}\beta d^4,$$

where g is gravitational acceleration, α the thermal expansion, κ the thermometric conductivity, and ν the kinematic viscosity. The parameter β is the vertical temperature gradient, and d the depth of the fluid layer.

The dimensionless governing system of equations on the domain

$$\Omega = \{(x, y, z) : 0 \leq z \leq 1\}$$

can be written as

$$\begin{aligned} \frac{1}{P}\left\{\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\cdot\nabla)\,\mathbf{v}\right\} &= -\nabla p + \sqrt{R}\theta\mathbf{e}_z + \Delta\mathbf{v} \\ \frac{\partial\theta}{\partial t} + (\mathbf{v}\cdot\nabla)\,\theta &= \sqrt{R}v_3 + \Delta\theta \\ \operatorname{div}\,\mathbf{v} &= 0, \end{aligned}$$

where P is the Prandtl number, $\mathbf{v} = (v_1, v_2, v_3)$ the fluid velocity, p the pressure, θ the deviation of temperature from the pure conduction state, and $\mathbf{e}_z$ the vertical unit vector. At the top and bottom bounding surfaces, two conditions are of interest: *rigid*, or no-slip, surfaces where $\mathbf{v}$ vanishes, and *free*, or no-stress, surfaces where $\partial v_1/\partial z = \partial v_2/\partial z = v_3 = 0$.

Rayleigh showed that instability in such a system manifests itself precisely when R exceeds a threshold value R_c, the critical Rayleigh number. This value, and its corresponding wave number k_c, take on different values for each of three distinct pairs of boundary conditions in the problem: the symmetric *free-free* and *rigid-rigid* cases; and the asymmetric *rigid-free* case. In [21], Rayleigh used this relationship to address the *free-free* case – for example, one in which a layer of fluid is suspended between two other fluids. In this case, he found the precise solution $R_c = 27\pi^4/4$, $k_c = \pi/\sqrt{2}$.

2.1 Rigid-rigid boundary conditions

Unlike the algebraic "eigenvalue" problem solved in [21], the characteristic equations relating R_c and k_c in the *rigid-rigid* case are transcendental. The first estimates for the *rigid-rigid* Rayleigh and wave numbers are due to Jeffreys [13] (1926) and Low [17] (1929), although the first rigorous treatment appears in 1940, from Pellew and Southwell [20]. The very first verifiable calculation for R_c and k_c did not appear until 1999. Using interval analysis techniques, Jeng and Hassard produced error-bounded estimates of the constants to fifty-digit precision [14].

2.2 Asymmetric boundary conditions

In this paper, we focus our attention on precise calculations of the Rayleigh number and wave number for the asymmetric, or *rigid-free*, Rayleigh-Bénard problem. The asymmetric problem – Bénard's original physical experiment – is particularly compelling for several reasons. As in its *rigid-rigid* counterpart, the characteristic equations governing the *rigid-free* case are transcendental, and to date we have found no verifiable, error-bounded approximations for the corresponding constants R_c and k_c in the literature. Second, while solutions of the *rigid-rigid* case observe an underlying $D_6 \dot{+} T^2 \times \mathbb{Z}_2$ symmetry, the lack of z-directional symmetry about the midplane in the asymmetric case leads to $D_6 \dot{+} T^2$ solution symmetry. While this inhomogeneity complicates analysis, it has been pointed out that the asymmetric boundary conditions may better describe planar dynamics in applications from geology, meteorology, astrophysics and physiology. See [3, 4, 6] and elsewhere.

In [11], the existence of a unique minimum Rayleigh number and unique corresponding wave number (k_c, R_c) for the asymmetric *rigid-free* problem was proved. In this paper, we construct error-bounded estimates for these constants, guaranteed to fifty-digit accuracy.

3 Formulation of the Asymmetric Problem

Using the Boussinesq approximation to the Navier-Stokes equations, Chandrasekhar [7] derived the neutral stability equation $F\left(k,R\right)=0$ for the *rigid-free* Rayleigh-Bénard problem. This equation was restated in the following form in [14], and later in [11]:

$$F\left(k,R\right)=G\left(k,R\right)+H\left(k,R\right)=0,\ \text{where} \tag{1}$$

$$G\left(k,R\right)=\begin{cases}-\theta\coth(\theta/2) & \text{for } R<k^4,\\ -2 & \text{for } R=k^4,\\ -\phi\cot(\phi/2) & \text{for} R>k^4,\end{cases} \tag{2}$$

$$\theta\left(k,R\right)=2\sqrt{k^2-\left(k^2R\right)^{(1/3)}},$$
$$\phi\left(k,R\right)=2\sqrt{\left(k^2R\right)^{(1/3)}-k^2},$$

and

$$H\left(k,R\right)=\frac{(a+\sqrt{3}b)\sinh\left(a\right)-(\sqrt{3}a-b)\sin\left(b\right)}{\cosh\left(a\right)-\cos\left(b\right)}, \tag{3}$$

$$a\left(k,R\right)=\sqrt{2\sqrt{k^4+k^2\left(k^2R\right)^{1/3}+\left(k^2R\right)^{2/3}}+\left(2k^2+\left(k^2R\right)^{1/3}\right)},$$
$$b\left(k,R\right)=\sqrt{2\sqrt{k^4+k^2\left(k^2R\right)^{1/3}+\left(k^2R\right)^{2/3}}-\left(2k^2+\left(k^2R\right)^{1/3}\right)}.$$

The critical Rayleigh number R_c and critical wave number k_c represent the least nonzero Rayleigh number R, and the corresponding wave number k, satisfying the neutral stability equation $F(k,R)=0$ as given in Eq. (1).

3.1 Uniqueness of the critical Rayleigh number

Implicit plotting of the neutral stability curves $F(k,R)=0$ suggest a sequence of interlocking curves $\mathcal{R}_n(k)$. This notion can be made rigorous with the introduction of "separation curves" $S_j(k)$, as shown in [14]:

$$S_j\left(k\right)=\frac{\left(k^2+j^2\pi^2\right)^3}{k^2},\ j\in\mathbb{N}. \tag{4}$$

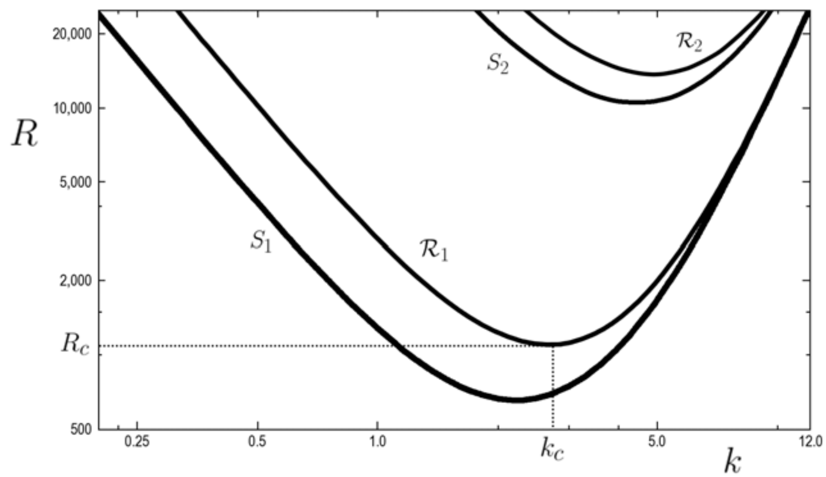

Figure 1: The sequence of reference curves $S_j(k)$ establish the existence of a "least" nonzero neutral stability curve $\mathcal{R}_1$ on which $F(k,R) = 0$. Approximate location of (k_c, R_c) shown; redrawn from MATLAB plot.

Observe from Eqs. (1), (2) that the curves $S_j(k)$ given in Eq. (4) represent precisely those points (k, R) on which the neutral stability function $F(k, R)$ fails to exist. We cite the following theorem from [11]:

Theorem 3.1. *On the region*

$$\mathcal{T} = \{(k, R) : k > 0,\ R > 0,\ R \neq S_j(k)\}, \tag{5}$$

for $F(k, R)$ given in Eq. (1):

1. *The function $F(k, R)$ is real analytic;*
2. *for fixed k, the partial derivative $\frac{\partial F}{\partial R}$ is strictly positive; and*
3. *the solution set for $F(k, R) = 0$ is of the form $\mathcal{R}_j(k)$, $j = 1, 2, \ldots,$ where for each $k > 0$, $\mathcal{R}_j$ is real analytic with $S_j(k) < \mathcal{R}_j(k) < S_{j+1}(k)$.*

In [11], this result was used to prove that the least neutral stability curve $\mathcal{R}_1(k)$ (see Figure 1) attains a unique minimum (k_c, R_c), where R_c denotes the critical Rayleigh number, and k_c its corresponding critical wave number. More precisely:

Theorem 3.2. *In the region $\mathcal{D}$, given by*

$$\mathcal{D} = \{(k, R_j(k)) :\ 2.5 < k < 3.1,\ 1.12 \leq j \leq 1.14\}, \tag{6}$$

where

$$R_j(k) = \frac{\left(k^2 + j^2\pi^2\right)^3}{k^2},$$

the implicit second derivative $\mathcal{R}_1''(k)$ *of* $\mathcal{R}_1(k)$ *is strictly positive, and the neutral stability curve* $\mathcal{R}_1(k)$ *for the asymmetric rigid-free Rayleigh-Bénard problem attains a unique minimum* (k_c, R_c)*.*

4 Main Results

In this paper, we attempt to isolate the unique least Rayleigh number R_c and corresponding wave number k_c with the property that $F(k_c, R_c) = 0$, for $F(k, R)$ as given in Eq. (1). To construct rigorous, error-bounded estimates for these constants, we appeal to interval analysis on the region $\mathcal{D}$ as given in Theorem 3.2.

4.1 Interval analysis

Broadly defined, interval analysis is a set of computational techniques which allow for calculation of 'computable' problems to arbitrary precision. More specifically, interval analysis admits variables of 'interval' type (e.g. $[a, b]$), allows interval-based evaluation of functions, and returns output of interval type. By consistent outward rounding at each step, interval analysis provides verifiable, error-bounded and arbitrarily precise computations. More substantial treatments can be found in [1, 18, 23] and elsewhere.

Using interval analysis, our calculation of the critical constants (k_c, R_c) is in fact a calculation of a closed rectangle $[k_L, k_U] \times [R_L, R_U]$ that is guaranteed to contain (k_c, R_c). The width of the k- and R-intervals, $k_U - k_L$ and $R_U - R_L$ respectively, then determine the precision with which the critical numbers (k_c, R_c) can be verifiably reported.

Theorem 4.1. *The critical Rayleigh number* R_c *and corresponding critical wave number* k_c *for the rigid-free Rayleigh-Bénard problem take the values:*

$$\begin{array}{lcrllllll} R_c & = & 1100. & 6496068876 & 7678462749 & 1888847271 & 1957506710 & 5545241401 & 2; \\ k_c & = & 2. & 6823217576 & 9341424484 & 3892930110 & 7059823340 & 0144144019 & 932, \end{array}$$

with an error of at most one digit in the last decimal place.

Proof. The critical values (k_c, R_c) must lie in the region $\mathcal{D}$ as given in Eq. (6). It is convenient to consider the search region $\mathcal{E}$:

$$\mathcal{E} = \left\{(k, R) : 2.5 \le k \le 3.1, 1034 \le R \le 1176\right\},$$

noting that $\mathcal{E} \supset \mathcal{D}$, and therefore that $(k_c, R_c) \in \mathcal{E}$. We considered subrectangles $\mathcal{C} \subseteq \mathcal{E}$ in an attempt to rule out candidate regions where (k_c, R_c) might lie; this exclusion would then leave a smaller region $\mathcal{E} - \cup\mathcal{C}$ guaranteed to contain the values (k_c, R_c).

Note that it is necessarily the case that $F(k_c, R_c) = 0$, so any subrectangle $\mathcal{C} \subseteq \mathcal{E}$ with $0 \notin F(\mathcal{C})$ can be eliminated as a region which might contain (k_c, R_c). Similarly, on any region containing (k_c, R_c), the implicit derivative $\mathcal{R}_1'(k)$ must vanish at some point; therefore any subrectangle $\mathcal{C} \subseteq \mathcal{E}$ with $0 \notin \mathcal{R}_1'(k)$ on $\mathcal{C}$ can also be eliminated as a candidate region for (k_c, R_c).

Observe that implicit differentiation of the neutral stability equation $F(k, R) = 0$ (see Eq. (1)) gives

$$\mathcal{R}_1'(k) = -\frac{\frac{\partial F}{\partial k}}{\frac{\partial F}{\partial R}}. \tag{7}$$

Since $\frac{\partial F}{\partial R}$ is strictly positive for fixed k in the region $\mathcal{T}$ as given in Eq. (5), and since $\mathcal{E} \subset \mathcal{T}$, it follows that $0 \notin -\frac{\partial F}{\partial k}(\mathcal{C})$ implies that $0 \notin \mathcal{R}_1'(k)$ on $\mathcal{C} \subseteq \mathcal{E}$.

Running GNU MPFI (Multiple Precision Floating Point Interval) Library on an IBM zEnterprise BladeCenter Extension, we employed interval analysis to evaluate $F(k, R)$ and $-\frac{\partial F}{\partial k}(k, R)$ on various subrectangles $\mathcal{C} \subseteq \mathcal{E}$, beginning with the rectangle $\mathcal{C} = \mathcal{E}$ itself. As expected, we found $0 \in F(\mathcal{E})$ and $0 \in -\frac{\partial F}{\partial k}(\mathcal{E})$. We then divided $\mathcal{E}$ into four equal, closed subrectangles $\mathcal{C}_i$, $i = 1, \ldots, 4$, of half the width and half the length of $\mathcal{E}$. On these subrectangles we evaluated $F(\mathcal{C}_i)$ and $-\frac{\partial F}{\partial k}(\mathcal{C}_i)$, looking for regions $\mathcal{C}_i$ where either $0 \notin F(\mathcal{C}_i)$, or $0 \notin -\frac{\partial F}{\partial k}(\mathcal{C}_i)$, and thereby refining the (k_c, R_c) search region to $\mathcal{E} - \mathcal{C}_i$.

We continued the process of re-division and re-evaluation of the neutral stability function and its derivative on a sequence of subrectangles of decreasing size: first $\mathcal{C}_{i,j}$, $i, j = 1, \ldots, 4$; then $\mathcal{C}_{i,j,k}$, $i, j, k = 1, \ldots, 4$; et cetera, eliminating as we proceeded those subrectangles that could be ruled out as containing (k_c, R_c). See Figure 2. In total, we evaluated 491,652 subrectangles $\mathcal{C} \subset \mathcal{E}$, until the refined (k_c, R_c) search region:

$$\mathcal{H} = \mathsf{co}\left\{\mathcal{E} - \left(\left(\bigcup_{0 \notin F(\mathcal{C})} \mathcal{C}\right) \bigcup \left(\bigcup_{0 \notin -\frac{\partial F}{\partial k}(\mathcal{C})} \mathcal{C}\right)\right)\right\}, \tag{8}$$

was calculated to a width less than or equal to 10^{-51} on each side. Here, co denotes the rectangular convex hull of the expression. The rectangle $\mathcal{H}$ is precisely the interval $[k_L, k_U] \times [R_L, R_U]$:

R_L	=	1100.	6496068876	7678462749	1888847271	1957506710	5545241401	13
R_U	=	1100.	6496068876	7678462749	1888847271	1957506710	5545241401	32
k_L	=	2.	6823217576	9341424484	3892930110	7059823340	0144144019	9317
k_U	=	2.	6823217576	9341424484	3892930110	7059823340	0144144019	9334,

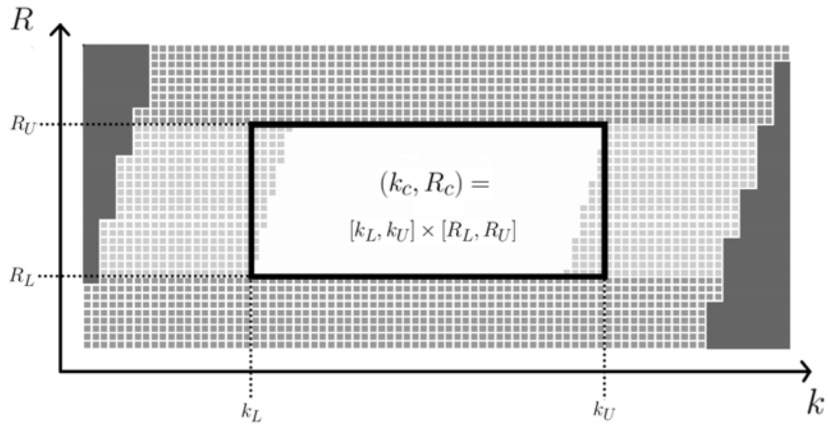

Figure 2: Close-up of subrectangles $\mathcal{C}$ near (k_c, R_c) after final iteration of inclusion tests. Black box (center) indicates the position of $(k_c, R_c) = [k_L, k_U] \times [R_L, R_U]$, given by the rectangular convex hull $\mathcal{H}$ in Eq. (8). Darker gray grid indicates subrectangles $\mathcal{C}$ rejected because $0 \notin F(\mathcal{C})$. Lighter gray grid indicates subrectangles $\mathcal{C}$ rejected because $0 \notin \mathcal{R}'(k)$ on $\mathcal{C}$. Solid dark regions on left- and right-hand sides indicate rectangles eliminated during previous iterations. Grid size is approximately $3.9 \cdot 10^{-55}$ in the k-direction, and $2.7 \cdot 10^{-54}$ in the R-direction.

in which (k_c, R_c) is guaranteed to lie. $\square$

4.2 Verification of results

Once $(k_c, R_c) = [k_L, k_U] \times [R_L, R_U]$ has been isolated, it is a straightforward task to verify the computations used to find it. By Theorem 3.1, there exists $\epsilon_R > 0$ such that the neutral stability function $F(k, R)$ given in Eq. (1) is strictly negative on the rectangle Q_1:

$$Q_1 = [k_L, k_U] \times [R_L - \epsilon_R, R_L]\,;$$

and strictly positive on the rectangle Q_3:

$$Q_3 = [k_L, k_U] \times [R_U, R_U + \epsilon_R]\,.$$

Further, by Theorems 3.1 and 3.2 and Eq. (7), there exists $\epsilon_k > 0$ such that the quantity $-\frac{\partial F}{\partial k}$ is strictly positive on the rectangle Q_2:

$$Q_2 = [k_U, k_U + \epsilon_k] \times [R_L, R_U]\,;$$

and strictly negative on the rectangle Q_4:

$$Q_4 = [k_L - \epsilon_k, k_L] \times [R_L, R_U]\,.$$

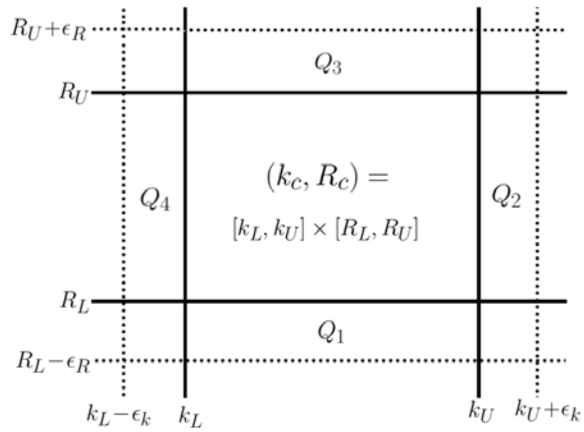

Figure 3: The rectangle $(k_c, R_c) = [k_L, k_U]\times[R_L, R_U]$, with bounding rectangles $Q_1, \ldots, Q_4$. The value for the parameter ϵ_k is slightly larger than $3.9 \cdot 10^{-55}$; ϵ_R is slightly larger than $2.6 \cdot 10^{-54}$. Figure not to scale.

The bounding rectangles $Q_i, i = 1, \ldots, 4$, are pictured in Figure 3, with size exaggerated for clarity. Again using interval analysis, we evaluated $F(Q_1)$, $F(Q_3)$, $-\frac{\partial F}{\partial k}(Q_2)$, and $-\frac{\partial F}{\partial k}(Q_4)$:

$$\begin{aligned} F(Q_1) &< -2.0 \cdot 10^{-54} \\ F(Q_3) &> 1.7 \cdot 10^{-54} \\ -\frac{\partial F}{\partial k}(Q_2) &> 2.9 \cdot 10^{-53} \\ -\frac{\partial F}{\partial k}(Q_4) &< -2.9 \cdot 10^{-53}. \end{aligned} \tag{9}$$

Were it the case that $(k_c, R_c) \notin [k_L, k_U] \times [R_L, R_U]$, then at least one of the calculations $F(Q_1)$, $F(Q_3)$, $-\frac{\partial F}{\partial k}(Q_2)$, and $-\frac{\partial F}{\partial k}(Q_4)$ would fail to attain its expected sign.

5 Discussion

Theorem 4.1 provides a verifiable, error-bounded calculation for the critical Rayleigh number and wave number for the *rigid-free* Rayleigh-Bénard problem, accurate to fifty-digit precision. In [14], Jeng and Hassard remarked that in bifurcation calculations on the solution set, "highly accurate values [for k_c and R_c] will be needed to compensate for a loss of precision during the subsequent computation."

It is interesting to compare the result of Theorem 4.1:

$$R_c = 1100.649606887676784627491888847271195750671055452414 01,$$

to historical approximations, by year, of the critical Rayleigh number. See Table 1.

Table 1: Selected approximations of the critical Rayleigh number R_c for the *rigid-free* Rayleigh-Bénard problem.

Year	Author	R_c	Notes
1929	Low [17]	1108.	subsequent text in [17]: '*nearly* 1108'
1940	Pellew and Southwell [20]	1100.65	original notation: '$17,610.5/16 = 1100.65$'
1958	Reid and Harris [22]	no decimal given[a]	estimate for $16R_c$ given as $17,610.39$
1961	Chandrasekhar [7]	1100.65	original notation: '$17,610.39/16 = 1100.65$'; author cites [22]
1993	Koschmieder [16]	1100.7	
1998	Getling [10]	1100.657	
2004	Drazin and Reid [9]	1101.	original notation: '$17,610.39/2^4 = 1101$'; authors cite [22]

[a]Reid and Harris [22] do not provide an explicit decimal expression for R_c. This may explain how two different values, given in [7] and [9], are each attributed to [22].

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